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Teknik Informatika - STEI ITB

Object Class Feature

Pemrograman Berorientasi Objek

Compare (ver 1)

```
public static void compare1() {
        Object obj1 = new Object();
        Object obj2 = new Object();
        Object obj3 = obj1;

        System.out.println(obj1==obj2);
        System.out.println(obj1==obj3);
        System.out.println(obj1.equals(obj2));
        System.out.println(obj1.equals(obj3));
        System.out.println(obj1.hashCode());
        System.out.println(obj2.hashCode());
        System.out.println(obj3.hashCode());
}
```

Output (ver 1)

```
false
true
false
true
746292446
1072591677
746292446
```

Compare (ver 2)

```
public static void compare2() {
        String st1 = new String("Halo");
        String st2 = new String("Halo");
        String st3 = st1;

        System.out.println(st1==st2);
        System.out.println(st1==st3);
        System.out.println(st1.equals(st2));
        System.out.println(st1.equals(st3));
        System.out.println(st1.hashCode());
        System.out.println(st2.hashCode());
        System.out.println(st3.hashCode());
}
```

Output (ver 2)

```
false
true
true
true
2241628
2241628
2241628
```

Compare (ver 3)

```
public static void compare3() throws CloneNotSupportedException {
        Point p1 = new Point(5,10);
        Point p2 = new Point(5,10);
        Point p3 = p1;

        System.out.println(p1==p2);
        System.out.println(p1==p3);
        System.out.println(p1.equals(p2));
        System.out.println(p1.equals(p3));
        System.out.println(p1.hashCode());
        System.out.println(p2.hashCode());
        System.out.println(p3.hashCode());

        p3 = (Point) p1.clone();
        System.out.println(p1);
        System.out.println(p3);
        System.out.println(p1==p3);
        System.out.println(p1.equals(p3));
        System.out.println(p3.hashCode());
}
```

Class Point

```
public class Point implements Cloneable {
        private int x = 0;
        private int y = 0;
        // a constructor!
        public Point(int x, int y) {
                this.x = x;
                this.y = y;
        }
        public int getX() {
                return x; }
        public int getY() {
                return y; }
        public void setX(int ax) {
                x = ax; }
        public void setY(int ay) {
                y = ay; }

        @Override
        public String toString() {
                return "Point [x=" + x + ", y=" + y + "]";
        }
```

Class Point (...)

```
@Override
public boolean equals(Object obj) {
        if (this == obj)
                return true;
        if (obj == null)
                return false;
        if (getClass() != obj.getClass())
                return false;
        Point other = (Point) obj;
        if (x != other.x)
                return false;
        if (y != other.y)
                return false;
        return true;
}
```

Class Point (...)

```
        @Override
        public int hashCode() {
                final int prime = 31;
                int result = 1;
                result = prime * result + x;
                result = prime * result + y;
                return result;
        }

        @Override
        protected Object clone() throws CloneNotSupportedException {
                return super.clone();
        }

}
```

Output (ver 3)

```
false
true
true
true
1126
1126
1126
Point [x=5, y=10]
Point [x=5, y=10]
false
true
1126
```

Compare (ver 4)

```
public static void compare4() throws CloneNotSupportedException {
        Point p = new Point(10,20);
        Rectangle r1 = new Rectangle(p,30,40);

        System.out.println(p);
        System.out.println(r1);

        Rectangle r2 = (Rectangle) r1.clone();
        System.out.println(r2);
        System.out.println(r1==r2);
        System.out.println(r1.equals(r2));
        System.out.println(r1.getOrigin()==r2.getOrigin());
        System.out.println(r1.hashCode());
        System.out.println(r2.hashCode());
}
```

Class Rectangle

```
public class Rectangle implements Cloneable {
        public int width = 0;
        public int height = 0;
        public Point origin;
        // four constructors
        public Rectangle() {
                origin = new Point(0, 0); }
        public Rectangle(Point p) {
                origin = p; }
        public Rectangle(int w, int h) {
                origin = new Point(0, 0);
                width = w; height = h; }
        public Rectangle(Point p, int w, int h) {
                origin = p; width = w; height = h; }
        // a method for moving the rectangle
        public void move(int x, int y) {
                origin.setX(x);
                origin.setY(y);
        }
```

Class Rectangle (…)

```
@Override
public boolean equals(Object obj) {
        if (this == obj)
                return true;
        if (obj == null)
                return false;
        if (getClass() != obj.getClass())
                return false;
        Rectangle other = (Rectangle) obj;
        if (height != other.height)
                return false;
        if (width != other.width)
                return false;
        if (origin == null) {
                if (other.origin != null)
                        return false;
        } else if (!origin.equals(other.origin))
                return false;
        return true;
}
```

Class Rectangle (…)

```
        @Override
        public int hashCode() {
                final int prime = 31;
                int result = 1;
                result = prime * result + height;
                result = prime * result + width;
                result = prime * result + ((origin == null) ? 0 :
origin.hashCode());
                return result;
        }

//      @Override
//      protected Object clone() throws CloneNotSupportedException {
//              return super.clone();
//      }

        @Override
        protected Object clone() throws CloneNotSupportedException {
                Rectangle tmp = (Rectangle) super.clone();
                tmp.origin = (Point) origin.clone();
                return tmp;
        }
}
```

Output (ver 4)

```
Point [x=10, y=20]
Rectangle [width=30, height=40, origin=Point [x=10, y=20]]
Rectangle [width=30, height=40, origin=Point [x=10, y=20]]
false
true
false
70452
70452
```

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